Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A Wearable Transcranial Doppler System			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio·Medical		Bio·Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Ultrasound Diagnostic Devices	Medical Information & Systems	Other Medical Information & Systems
Concept of Proposed Technology	Cerebrovascular diseases, including stroke, traumatic brain injury, and perioperative cerebral ischemia, represents a significant global health challenge, requiring continuous and reliable monitoring of cerebral hemodynamics[1-2]. Although transcranial Doppler (TCD) ultrasound is a clinically established modality for the real-time evaluation of cerebral blood flow, its widespread and long-term use remains limited by operator dependency, and bulky cart-based systems[3]. These limitations are particularly critical in intensive care units, neurosurgical wards, and perioperative monitoring, where continuous and automated cerebral monitoring is essential.[4] Conventional TCD systems requires proficient clinicians to locate the target vessels or has bulky form factor for robotized scanning, disadvantageous for wearable form factor. Such methods are also qualitative as the relative orientation between the TCD probe and target vessels are blinded. This project focuses on the development of a next-generation wearable TCD system designed to overcome the limitations of conventional cart-based ultrasound imaging platforms and conventional TCD systems. The proposed technology employs an adhesive, patch-type, and electrically-scannable cross phased array ultrasound transducer positioned over the temporal bone window to enable real-time, non-invasive, and long-term monitoring of hemodynamic parameters in major brain vasculatures (e.g., Circle of Willis, middle/anterior/posterior cerebral arteries). In addition, automated Doppler gate			

Rationale for Joint R&D Project	localization and motion artifact detection algorithms are implemented to ensure robust and reliable measurements during prolonged use. The Korean partner contributes strong expertise in miniaturized ultrasound imaging systems, enabling the development of an adhesive, patient-comfortable TCD patch device optimized for longitudinal cerebral monitoring. In parallel, the U.S. partner provides complementary strengths with profound physiological insights with large pre-clinical animal models of human brain and proximity to world-leading clinical research capabilities.
Key Words	1. Wearable transcranial Doppler system 2. Patch-type phased array transducer 3. Automated Doppler gate localization 4. Motion artifact detection

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	Development of a next-generation wearable transcranial Doppler (TCD) system for continuous monitoring of cerebral blood flow			
R&D Target Specifications	• Wearable TCD system • Patch-type cross phased array • Automated Doppler gate localization • Motion artifact detection algorithms • Spectral Doppler accuracy >95% compared to ground-truth.			
R&D Plan	**Aim 1. Wearable TCD Prototyping** The project will build an electrically-scannable, economic, and wearable TCD prototype by implementing targeted hardware and system-level refinements to improve robustness, usability, and long-term monitoring performance. The patch-type cross phased array transducer and its fixation mechanism to enhance acoustic coupling at the temporal window and improve patient comfort during prolonged use. In parallel, system electronics and data acquisition parameters will be optimized to support stable continuous Doppler measurements under realistic conditions, including movement and variable head anatomies. **Aim 2. Automated Doppler Gate Localization** To reduce operator dependency and improve measurement reproducibility, the project will develop automated Doppler gate localization algorithms for major vasculatures under the skull layer (e.g., middle/anterior/posterior cerebral arteries). These algorithms will leverage Doppler signal characteristics, flow directionality, and temporal stability to automatically identify optimal sampling locations within cerebral arteries. Real-time implementation will be pursued to enable adaptive gate adjustment during long-term monitoring, ensuring consistent hemodynamic measurements despite changes in probe position or patient posture. **Aim 3. Motion Artifact Detection and Signal Quality Assessment** Motion-induced artifacts remain a key challenge for wearable cerebral ultrasound systems. To address this, the project will develop motion artifact detection algorithms that identify and classify signal degradation caused by head movement, probe displacement, or external disturbances. Signal quality metrics and intelligent filtering strategies will be incorporated to either suppress corrupted data segments or trigger automatic reacquisition, thereby enhancing the reliability of continuous Doppler monitoring.			

	Aim 4. Preclinical Modeling, Data Collection, and Validation The device will be applied to large-scale animals to measure the accuracy of the Doppler measurements of blood flow in a target vasculature, compared to the ground-truth measurement using epidural spectral Doppler at the same target using clinical ultrasound device. Statistical significance for improved accuracy will be also calculated compared to the conventional TCD device without electrical scanning. **Aim 5. Clinical/scientific Design Review** Clinical and scientific specifications of the TCD design (including measurable Doppler speed range, spatial scannable range, clinical usability, form factor, applicability to human anatomy, interface) will be reviewed by field clinicians and physiologists.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The U.S. partner is primarily responsible for the review of the proposed device in a global standard, and provides preclinical validation of the wearable TCD system, leveraging infrastructures and multidisciplinary expertise in relevant fields, including, but not limited to physiology, pre-clinical model for brain research, neurocritical care, and vascular surgery. Specifically, the U.S. partner should lead the design and execution of animal studies to assess the performance, robustness, and usability of the refined the wearable TCD system. The preclinical model is expected to allow direct comparison of the proposed PCD system with ground-truth Doppler measurements and conventional clinical TCD system(s). The U.S. partner will also provide clinical and scientific feedback and expertise to inform iterative system refinement, ensuring alignment with clinical workflows and unmet needs in neurocritical care and vascular surgery. Through close coordination with the Korean partner, the JHU will contribute to joint data analysis, interpretation of clinical findings, and dissemination of results through co-authored publications and international conferences.
Expected R&R of Korean SME Partner	The Korean team is primarily responsible for the technical development and system integration of the wearable TCD platform, building upon the existing prototype and advancing it toward a clinically robust and wearable system. Specifically, the Korean team will lead the refinement of the wearable TCD hardware, including modifications to the patch-type cross phased array transducer, fixation mechanism, and system electronics to enhance acoustic coupling, signal stability, and patient comfort during long-term monitoring. The team will also optimize data acquisition and embedded processing to support continuous and reliable Doppler measurements in wearable settings. In addition, the Korean team will develop and implement automated Doppler gate localization algorithms to reduce operator dependency and improve measurement reproducibility. Motion artifact detection and signal quality control algorithms will be designed to identify and mitigate signal degradation caused by patient movement or probe displacement, ensuring robust performance under clinical conditions. The Korean team will be responsible for full system integration, combining refined hardware and intelligent automation algorithms into a unified wearable TCD platform. Close collaboration with the U.S. partner will be maintained to incorporate clinical/scientific feedback into iterative system improvements. The Korean partner will also lead joint data analysis, algorithm refinement, and dissemination of technical outcomes through publications, patents, and international conferences, supporting the translational advancement of the jointly developed technology.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project will achieve a fully integrated, TCD system that combines a patch-type cross phased array ultrasound device with automated Doppler gate localization and motion artifact detection algorithms. By enabling continuous, non-invasive, and operator-independent cerebral hemodynamic monitoring, the technology represents a significant advancement beyond conventional cart-based TCD systems and existing wearable ultrasound approaches. In addition, the project will establish a scalable platform for long-term cerebral Doppler data acquisition, validated in real pre-clinical environments. The integration of miniaturized hardware with intelligent signal processing and automation algorithms provide the technical foundation for future AI-driven cerebrovascular diagnostics and next-generation wearable ultrasound systems. • Commercialization Potential The developed wearable TCD system has strong commercialization potential as a novel medical device for neurocritical care, neurosurgery, vascular surgery, and perioperative monitoring. Its compact form factor, ease of use, and reduced operator dependency address key barriers to adoption of conventional TCD, expanding its applicability beyond specialized ultrasound laboratories to general hospital wards and intensive care units. Furthermore, the pre-clinical datasets generated through this project enable the development of software-based diagnostic and monitoring solutions. This combination of hardware and data-driven software creates multiple commercialization pathways, including device sales, software licensing, and partnerships with medical device and digital health companies. • Industrial and Economic Impact Industrially, the project is expected to strengthen the competitiveness of Korea–US medical ultrasound and digital health industries by establishing leadership in wearable cerebral monitoring technologies. The joint development and validation framework accelerates translational readiness, supports regulatory pathways, and lowers barriers to global market entry. Economically, the proposed technology has the potential to reduce healthcare costs by enabling early detection of cerebral hemodynamic abnormalities and supporting patient management. By fostering cross-border collaboration, technology transfer, and high-value job creation in medical device and AI sectors, the project contributes to sustainable economic growth and reinforces the strategic partnership between Korea and the United States.

6. References

(1) W. J. Powers et al., “Guidelines for the early management of patients with acute ischemic stroke,” *Stroke*, vol. 50, no. 12, pp. e344-e418, 2019

(2) J. P. Fanning et. al., “Perioperative stroke,” *Nat Rev. Dis. Primers*, vol. 10, no. 3, 4072, 2024.

(3) K. Clare et al., “Safety and efficacy of a novel robotic transcranial Doppler,” *Scientific Reports*, vol. 11, 2022.

(4) A. R. Maas et al., “Traumatic brain injury: integrated approaches to improve prevention, clinical care, and research,” *Lancet Neurol.*, vol. 16, no. 12, pp. 987-1048, 2017.